ASSIGNMENT ON
Electromagnetic Induction
(EMI)

1) Magnetic flux 'ϕ' through a metal ring of resistance $R = 3\Omega$ varies with time according to relation $\phi = 3(at^3 - bt^2)$ Wb where $a = 2$ & $b = 6$. Determine maximum current induced in the ring during the interval from $t = 0$ to $t = 2$ sec.

[$A_{m}|_{max} = 6A$]

2) A coil having 500 square loops each of side 10cm is placed normal to a magnetic field which increases at the rate of 1T/s. Find induced emf? [$e = 5V$]

3) A coil having an area $2m^2$ is placed in a magnetic field which changes from $1\ Wb/m^2$ to $4\ Wb/m^2$ in an interval of 2sec. Find emf induced in coil?

[$e = 3V$]

4) In a magnetic field of 0.05 T, ~~area~~ area of a coil changes from $101\ cm^2$ to $100\ cm^2$ without changing the resistance which is 2Ω. Find the amount of charge that flows during this period. [$q = 2.5 \times 10^{-6} C$]

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5. A coil has an area of $0.05\ m^2$ and it has 800 turns. It is placed perpendicularly in a magnetic field of strength $4 \times 10^{-5}\ Wb/m^2$, it is rotated through $90°$ in 0.1 sec. Find the ~~average~~ average emf induced in the coil.

[0.016 volt]

6. A metal rod of length 2m is rotating with an angular velocity of 100 rad/sec in a plane perpendicular to a uniform magnetic field of 0.3T. Find potential difference between the ends of the rod.

[60V]

7. A metallic disc of area 'A' is rotating with a revolution frequency 'n' at right angle to a uniform magnetic field 'B'. Find emf induced betn the centre and the rim/circumference of the disc. [$e = BAn$]

8. An aeroplane in which the distance between the tips of the wing = 50m is flying horizontally from East to West with the speed of 360 km/hr over a place, where the vertical component of earth's magnetic field is $2 \times 10^{-4}\ Wb/m^2$. Find potential difference betn. the tips of the wing.

(1 volt)

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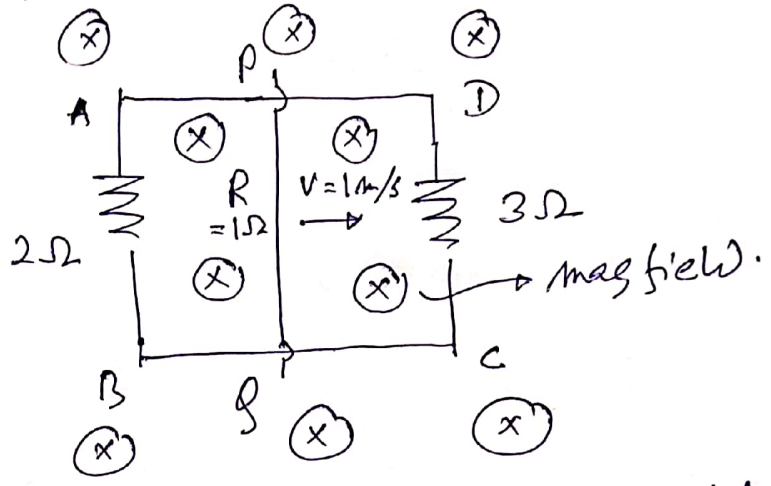
9. A rectangular loop with a sliding connector PQ of length 10cm is in a constant magnetic field of 0.1T perpendicular to the plane of loop. Resistance of rod PQ is $R = 1\Omega$ & those of sides AB and CD are 2Ω and 3Ω resp. a) Find current in connector rod PQ during its motion with constant velocity 1m/s ?



b) Which end of connector PQ is at higher potential.

c) And what will be the direction of induced current through resistors 'AB' & 'DC'.

[a) $i = \frac{1}{220}$ A, b) end P c) A → B & D → C]

10. A magnetic field $B = 2t + 4t^2$ Tesla, [t = time in sec] is perpendicular to a metallic ring of Area $2m^2$ & of resistance $R = 5\Omega$. Find charge flown in the ring in 0 to 2 sec.

[q = 8 Coulomb]

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